

Research Paper

Integrating hyperspectral radiation transfer modeling and deep transfer learning to estimate nitrogen density in winter wheat canopies

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ABSTRACT

Nitrogen is a core element that regulates winter wheat yield and quality, and its precise monitoring is crucial for sustainable agricultural development. Traditional empirical methods and physical inversion models face challenges in estimating canopy nitrogen density (CND), including weak generalization capabilities, ill-posed inversion, and heavy reliance on large amounts of labeled data. To address these issues, this study proposes a novel approach that integrates physical mechanisms with data-driven methods. It combines the N-PROSAIL radiative transfer model with deep transfer learning to achieve high-precision, robust estimation of CND in winter wheat. This study uses a large-scale, label-accurate simulated spectral dataset generated by the N-PROSAIL model as the source domain. This dataset is used to pretrain a one-dimensional convolutional neural network (SCAR-CNN) with dual attention mechanisms to learn the universal mapping relationship between spectral data and nitrogen content. A small field-measured dataset is subsequently employed to fine-tune the pretrained model through transfer learning, bridging the domain gap between the simulated and real data. The results from a two-year field trial demonstrate that the hybrid framework achieves significantly higher estimation accuracy ($R^2 = 0.81$, $RMSE = 0.14$) than traditional lookup table (LUT) inversion and untrained deep learning models, effectively mitigating the overfitting issue in small-sample scenarios. The model demonstrated outstanding generalizability and stability across growth stages, nitrogen gradients, and from ground-based to UAV platforms. This confirms the immense potential of integrating physical prior knowledge with deep transfer learning to overcome heterogeneity in agricultural remote sensing data and enhance model transferability, providing an interpretable and efficient technical approach for precise monitoring of crop nitrogen nutrition.

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1. Introduction

Nitrogen (N), as a key element constituting plant proteins, enzymes, and photosynthetic pigments, serves as a core nutrient that regulates winter wheat growth and development, grain quality, and yield (Diacono et al., 2013). Its availability directly affects photosynthetic efficiency and assimilate accumulation while also determining nitrogen transport capacity, ultimately influencing yield potential and environmental sustainability. However, nitrogen management faces significant challenges: insufficient nitrogen application severely limits yield and quality (Chlingaryan et al., 2018), whereas excessive application not only wastes resources and increases costs but also poses ecological

risks (Padilla et al., 2018). Therefore, precise and dynamic monitoring of winter wheat canopy nitrogen density (CND) is a prerequisite for achieving timely and variable fertilization (Cilia et al., 2014), optimizing the nitrogen nutrition index (NNI) (Lemaire et al., 2008), increasing nitrogen use efficiency (NUE), and minimizing environmental burdens.

Traditional nitrogen monitoring methods rely primarily on destructive sampling and laboratory analysis and exhibit inherent limitations, such as high costs, poor timeliness, and limited spatial coverage. These approaches struggle to meet the monitoring demands of modern agriculture at the field scale (Zhao et al., 2012). In contrast, hyperspectral remote sensing, with its nondestructive, rapid acquisition of vegetation reflectance spectra and ability to correlate these with biochemical components, has emerged as an ideal technological approach for large-scale, real-time assessments of crop nitrogen status (Ballester et al., 2017; Fu et al., 2020). To achieve accurate and robust estimation of vegetation parameters on the basis of spectral data, researchers have developed

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several modeling approaches, including empirical and physical models. Empirical models directly establish mapping relationships between spectral features and nitrogen content using statistical or machine learning algorithms (Camino et al., 2018; Verrelst et al., 2019). However, their performance relies heavily on training data, making them susceptible to environmental noise interference and exhibiting weak generalization capabilities. The performance often significantly declines when different crop varieties, growth stages, or geographical regions are considered (Wang et al., 2020). Physical models, which are grounded in radiative transfer theory, simulate light propagation within the canopy to generate reflected spectra. They estimate CND indirectly by inverting biochemical parameters (Féret et al., 2021; Wang et al., 2018). However, they face ill-posed inversion problems: the high dimensionality and interdependencies of input parameters (leaf area index, leaf inclination distribution, leaf structure, etc.) lead to non-unique solutions (Li et al., 2018; Sun et al., 2022). Thus, both empirical and physical models exhibit significant limitations in terms of nitrogen estimation.

In recent years, hybrid modeling approaches that combine the flexibility of empirical models with the mechanistic nature of physical models have emerged as key research directions for enhancing the accuracy and robustness of remote sensing estimates of vegetation parameters (Berger et al., 2020a; Camps-Valls et al., 2016; Verrelst et al., 2015). This approach leverages physical models to describe the radiation transport processes within the canopy, generating simulated spectral data with clear physical meaning. Moreover, it employs empirical algorithms to flexibly establish mapping relationships between spectral data and target parameters, thereby maintaining high inversion efficiency (Dehghan-Shoar et al., 2023; Li et al., 2024). However, vegetation parameter inversion remains heavily reliant on large amounts of high-quality field-annotated data. Agricultural applications often face challenges such as scarce field samples, high annotation costs, and limited spatiotemporal representativeness, which constrain the model's promotion and application in practical production (Zhang et al., 2018). To address this, transfer learning has been introduced to the field. It works by transferring generalizable features learned from source domains (such as other crops, regional data, or simulated data). This approach effectively mitigates the challenges posed by insufficient ground truth data in the target domain. Consequently, it enhances the model's accuracy and robustness across different crop varieties and environmental conditions (Behera et al., 2021).

However, existing methods still face significant challenges in scenarios with sparse data and cross-domain tasks. Particularly when integrating physical mechanisms with transfer learning, existing hybrid modeling studies predominantly rely on simulation data from a single crop growth stage or limited parameter ranges for pre-training. This results in insufficient scene adaptability of pre-trained features, leading to failure during generalization across growth stages and nitrogen gradients due to parameter distribution shifts (Berger et al., 2020b). While some nitrogen estimation efforts incorporating transfer learning achieve cross-domain feature transfer, they lack constraints from RTM physical mechanisms. Feature learning becomes detached from canopy light transport processes, resulting in weak physical interpretability of retrieval outcomes (Fu et al., 2025). Other studies have integrated RTMs with deep learning to construct hybrid models, but these approaches often fail to specifically preserve physically relevant features, leading to overfitting when applied to field data (Zhang et al., 2024). Therefore, this study proposes a novel approach that integrates hybrid modeling with deep transfer learning (DTL). The core strategy involves the following steps: First, the PROSAIL model is used to couple leaf and canopy radiative transfer processes, generating a large-scale, accurately labeled simulated spectral and CND dataset covering typical biochemical parameters and canopy structures as the source domain. A deep neural network is pretrained on this simulated dataset to establish a universal mapping relationship from spectral data to CND, particularly when key spectral features related to nitrogen status are identified (Li

et al., 2019; Yang et al., 2015). DTL is subsequently employed to perform supervised fine-tuning of the pretrained model using a small amount of field-measured hyperspectral data and actual CND data from winter wheat (Zhang et al., 2021). Finally, the fine-tuned model undergoes systematic validation on an independent test dataset. This strategy fully leverages the physical generalizability of RTMs and the nonlinear modeling advantages of deep learning, effectively mitigating the risk of overfitting on small samples. Consequently, the model not only possesses physical interpretability but also maintains high computational efficiency. This provides an interpretable and efficient new paradigm for addressing the long-standing challenges of data heterogeneity and model transferability in agricultural remote sensing (Li et al., 2025a; Li et al., 2025b).

This study focuses on winter wheat to propose and validate a hybrid framework that integrates physical mechanisms with DTL, aiming to increase the accuracy and robustness of CND estimation for winter wheat. The specific objectives include (1) constructing a simulated dataset constrained by radiation transfer models (RTMs) for deep network pre-training combined with DTL for adaptive fine-tuning using small-sample field data and (2) evaluating the accuracy and generalizability of this hybrid framework in terms of estimating winter wheat CND across years, growth stages, and nitrogen gradients.

2. Materials and methods

2.1. Experimental design

The study area is located at the Jiaozuo Academy of Agricultural and Forestry Sciences in Henan Province, which is between 34°48'16" and 35°29'30" north latitude and 112°32'48" to 113°39'45" east longitude, with an average elevation of 82.5 m. The climate is a warm temperate continental monsoon climate, featuring distinct seasons. Precipitation is concentrated from June to September, with annual rainfall ranging from 600 to 700 mm. A two-year winter wheat cultivation experiment was conducted in the study area. In the first year (October 2023 to June 2024), 48 plots (1.4 m × 9.5 m per plot, with buffer strips between plots to eliminate cross-plot effects) were established to test four wheat varieties (Jiaomai 8, Lifeng 16, Lifeng 66, and Zhongmai 578) under four nitrogen fertilizer rates (N0: 0 kg/hm², N1: 186 kg/hm², N2: 373 kg/hm², and N3: 560 kg/hm²), with three replicates per treatment. Samples were collected at the jointing stage (April 6), heading stage (April 21), flowering stage (May 5), and grain-filling stage (May 19). In the second year (October 2024 to June 2025), 48 plots of identical size were established. The varieties were adjusted to Jiao Mai 8, Zhongmai 69, and Zhongmai 578. The nitrogen fertilizer treatments remained consistent throughout the first year, with each treatment being replicated four times. Samples were collected during the jointing stage (March 21), heading stage (April 16), flowering stage (April 28), and grain-filling stage (May 7). A total of 384 samples were collected over the two years (See Fig. 1).

2.2. Data acquisition

2.2.1. Ground-based hyperspectral imaging

In this study, an ASD FieldSpec4 hyperspectral radiometer (Analytical Spectra Devices, Inc., Boulder, CO, USA) was used to acquire spectral reflectance data from winter wheat canopies. The instrument has a spectral range of 350–2500 nm and a field of view of 25°. In the visible to near-infrared region (350–1000 nm), the sampling interval is 1.4 nm, with a spectral resolution of 3 nm; in the shortwave infrared region (1000–2500 nm), the sampling interval is 1.1 nm, with a spectral resolution of 8 nm. Data collection was strictly conducted on clear, windless, and cloudless days during the stable solar radiation period around local noon (10:00–14:00). During operation, the sensor probe was strictly maintained in a vertical downward orientation and was consistently positioned approximately 30 cm above the crop canopy. To mitigate

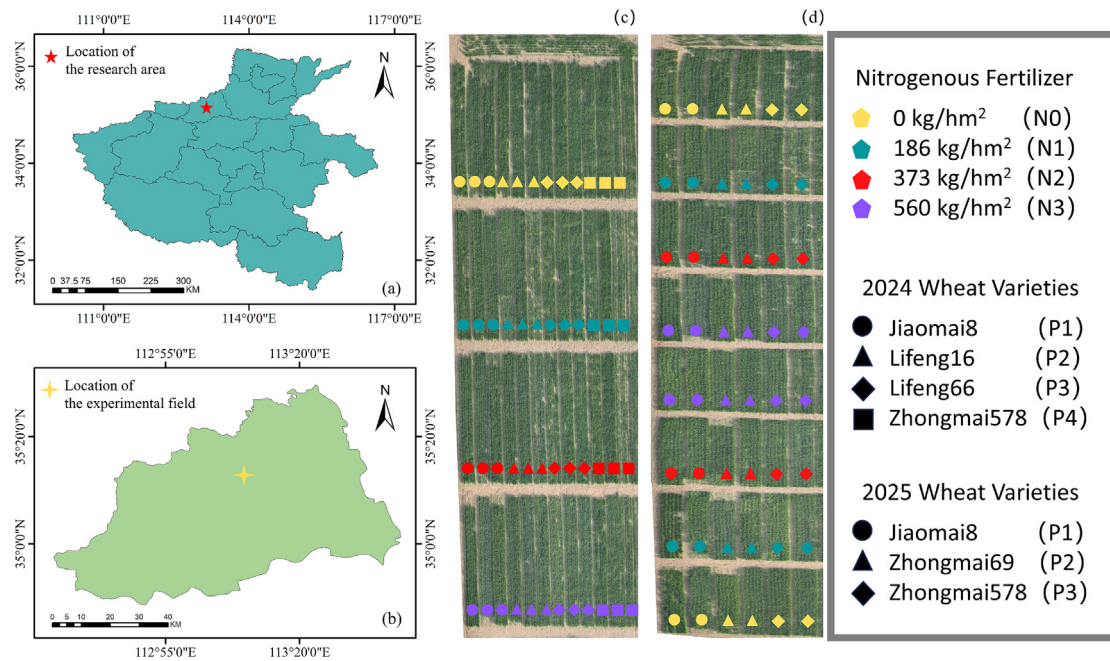


Fig. 1. Study Area (a: Province where the study area is located; b: Location of the study area; c and d: Experimental designs for the first and second years of the test field, respectively).

environmental light variations, a calibration using a standard reference panel was performed every 10 to 15 min or whenever significant fluctuations in light intensity occurred. Within each experimental plot, 30 spectral curves were uniformly acquired and averaged to constitute the plot's raw spectral data.

2.2.2. UAV hyperspectral imaging

In this study, a GaiaSky-mini3-VN unmanned aerial vehicle (UAV)-borne hyperspectral imaging system mounted on a DJI M350 RTK platform (DJI, Shenzhen, China) was used. The sensor's spectral sampling range was 400–1000 nm, with an average spectral resolution of 5 nm. In 224-channel mode, the spectral sampling interval was 2.7 nm, and the spatial channel count was 1024. Flight missions were conducted on clear, cloudless days during the critical growth stage of winter wheat (from jointing to grain filling). The fixed altitude (AGL) was 40 m, with a 12-s hovering exposure time. The heading overlap rate was 60%, and the sideways overlap rate was 60%. With a 16 mm lens, hyperspectral data with a spatial resolution of 0.03 m were acquired (single-frame resolution: 1024 × 1003; 12-bit depth). Radiometric calibration was performed using a gray standard reflectance tarp (50%) prior to flight. The raw data underwent reflectance calibration, atmospheric correction, full-band registration, and mosaicking to generate a digital orthophoto (DOM) of the wheat field. Spectral reflectance at the plot level for the winter wheat canopy was extracted using the region of interest (ROI) tool in ENVI software.

2.2.3. N-PROSAIL-based canopy spectral reflectance simulation dataset

In this study, the N-PROSAIL radiation transfer model was employed to construct a large-scale lookup table (LUT) for efficient inversion of winter wheat CND. This model couples the improved N-PROSPECT leaf model (Yang et al., 2015) with the SAIL canopy model (Verhoef, 1984). Unlike the traditional PROSAIL model, N-PROSPECT incorporates leaf nitrogen density (LND) as a key input parameter and replaces the original chlorophyll absorption coefficient (K_{Cab}) with the nitrogen absorption coefficient (K_N), thereby more directly characterizing nitrogen density. In this study, the prior distribution ranges and probability distributions of key physiological and biochemical parameters of winter wheat (including LND, leaf area index, and leaf dry matter content) were determined on the basis of field measurement data. A

Markov chain Monte Carlo (MCMC) method was subsequently employed to systematically generate 6000 parameter combinations (Dehghan-Shoar et al., 2023), simulating hyperspectral reflectance within the 400–2500 nm range to construct database mapping parameters to spectral responses (See Table 1).

2.2.4. Plant measurement

Representative plants were selected from each experimental plot. The samples were first subjected to high-temperature blanching at 105 °C for 30 min and then transferred to an 85 °C oven for drying to a constant weight (48–72 h). During this process, the fresh weight (FW) and dry weight (DW) of the leaves were measured using an electronic balance with a 0.01 g precision. The nitrogen concentration (N%) was determined using the Duma combustion method (Thermo Fisher Scientific, USA). The leaf area index (LAI) was measured using an LAI-2200C Plant Canopy Analyzer (LI-COR, USA) on the basis of canopy porosity principles. Five measurements were taken per plot, averaged, and used to calculate the leaf area (LA). The leaf nitrogen density (LND) was calculated as the leaf nitrogen concentration multiplied by the leaf dry matter content. CND was derived by multiplying the LAI by the LND (Li et al., 2019). The radiative transfer model (RTM) parameterization involved determining the leaf dry matter content (Cm) and equivalent water thickness (Cw) according to Eqs. (3) and (4) (Li et al., 2024).

Table 1

Nominal values and ranges of input parameters for the N-PROSAIL model.

Variable name	Abbreviation	Min value	Max value	Units
Leaf structure parameter	N	1	2.5	–
Leaf nitrogen density	LND	60	230	μg/cm ²
Carotenoid content	Car	1	20	μg/cm ²
Brown pigment	cbrown	0	0	–
Equivalent water content	Cw	0.001	0.02	cm
Dry matter content	Cm	0.001	0.02	g/cm ²
Leaf area index	LAI	0.1	7	–
Average leaf inclination angle	ALIA	30	70	Degrees
Hot spot parameter	HSPOT	0.01	0.5	–
Observer zenith angle	TTO	0	0	Degrees
Soil brightness factor	R _{soil}	0	1	–
Solar zenith angle	SZA	30	30	Degrees

$$LND = 10000 * N\% * DW / LA \quad (1)$$

$$CND = LND * LAI \quad (2)$$

$$Cm = 10000 * DW / LA \quad (3)$$

$$Cw = 10000 * (FW - DW) / LA \quad (4)$$

2.3. Data processing

Spectral data acquired by the ASDFieldSpec4 hyperspectral radiometer and UAV hyperspectral sensors were resampled to a 1 nm resolution. Savitzky–Golay filtering with a window width of 13 and cubic polynomial fitting was applied to each grid cell's spectral data for noise reduction and smoothing. Subsequently, mean filtering was applied using a 3 nm sliding window to sparsify the spectral data to 3 nm intervals, thus reducing data redundancy between adjacent bands and enhancing computational efficiency for subsequent analyses (Patel et al., 2023). Following the aforementioned preprocessing, a genetic algorithm (GA) was employed to screen hyperspectral sensitive bands for winter wheat canopy nitrogen content. Genetic algorithms were chosen for their ability to efficiently search the vast combinatorial space of hyperspectral bands and identify globally optimal nonlinear feature subsets. Key parameter settings for the genetic algorithm are presented in Table 2, where mutation and crossover probabilities were dynamically adjusted using exponential decay and exponential growth strategies, respectively. The algorithm employed the coefficient of determination (R^2) of the support vector machine regression (SVR) model as the fitness evaluation metric (Fig. 2) (Pullanagari et al., 2021).

2.4. Deep transfer learning model construction

This study addresses the challenge of nitrogen remote sensing inversion for winter wheat by proposing a hybrid approach that integrates deep learning with physical models. In the deep learning domain, a one-dimensional convolutional neural network (SCAR-CNN) model combining dual attention mechanisms and residual connections was constructed (Fig. 3). The model employs three residual blocks as its backbone network, mitigating gradient vanishing through skip connections. It innovatively incorporates a channel attention module (CAM) and a spectral attention module (SAM) to dynamically learn important feature weights from the feature channel and spectral dimensions, respectively (Hu et al., 2024; Xiao et al., 2024). The CAM generates channel weights through global average pooling and a fully connected layer. Specifically, the input feature matrix $Y = [y_1, y_2, \dots, y_x]$ undergoes global average pooling to yield the channel statistics vector Z_{Avg} . The statistical features for the i -th channel are as follows:

$$Z_i^{Avg} = \frac{1}{L} \sum_{j=1}^L y_{ij} \quad (5)$$

After two fully connected layers (ratio = 8), the weight vector Z^* is obtained through sigmoid normalization:

$$Z^* = \sigma(W_2 * \text{ReLU}(W_1 * Z^{Avg})) \quad (6)$$

The original features and weighted features are multiplied channelwise and then the residuals are summed. ReLU activation is applied to obtain the output:

$$Y_{CAM} = Y + Y * Z^* \quad (7)$$

The SAM uses parallel convolutional branches to capture spatiotemporal correlations. This module employs two parallel convolutional branches: One uses a 1×1 convolution (without domain interference) followed by sigmoid normalization to obtain the following:

$$A_s = \sigma(\text{ConvD}_{1 \times 1}(Y)) \quad (8)$$

Another approach uses a 1×3 convolution (covering the domain band) with ReLU activation to extract local features, yielding the following:

$$F_s = \text{ReLU}(\text{ConvD}_{1 \times 3}(Y)) \quad (9)$$

The output of the attention module is also obtained through residual connections:

$$Y_{SAM} = Y + A_s * F_s \quad (10)$$

This model effectively integrates local feature extraction with a global attention mechanism, significantly enhancing the nonlinear modeling capability between hyperspectral data and nitrogen density in winter wheat canopies. For the physical model, the root mean square error (RMSE) between the field-measured or remotely sensed spectral data and the simulated spectra in the lookup table (LUT) was calculated to select the optimal parameter combination as the inversion result (Dorigo et al., 2007).

Both models (the physics-based LUT inversion and the data-driven SCAR-CNN) exhibit distinct advantages and limitations. Therefore, this study introduced a transfer learning framework to the original SCAR-CNN architecture, establishing a two-stage training paradigm involving pretraining followed by fine-tuning (Fig. 3). This approach first pretrains a residual network with dual attention mechanisms on the source domain dataset (PROSAIL simulated data) to acquire general spectral feature representation capabilities. Through a parameter freezing strategy, it is subsequently fine-tuned on the target domain (real-world data). By fixing the weights of the residual convolutional feature extractor and fine-tuning the parameters of the high-level regression head, it retains the physical prior knowledge learned from the source domain while adapting to the specific distribution of the target domain. This approach aims to mitigate overfitting issues in small-sample scenarios and enhance the model's cross-domain generalization ability.

2.5. Accuracy assessment

In the quantitative assessment of model performance, the coefficient of determination (R^2) and root mean square error (RMSE) are widely adopted as core accuracy evaluation metrics. These two measures complement each other in quantifying the degree of agreement between predicted and observed values. R^2 quantifies the proportion of variance in the target variable that the model can explain. Its value ranges from [0,1], with values closer to 1 indicating stronger explanatory power over observed data variability and higher model goodness-of-fit. RMSE, meanwhile, reflects the absolute magnitude of prediction errors. By averaging residual squares and taking the square root, it restores the same units as the target variable, making error values more intuitively interpretable. RMSE is particularly sensitive to large errors in predictions; a smaller value indicates higher predictive accuracy of the model.

Table 2
GA parameters.

Parameter name	Parameterization
population_size	100
generations	500
Fitness function	SVR
mutation_scheduler	(0.8, 0.2, 0.01)
crossover_scheduler	(0.2, 0.8, 0.01)

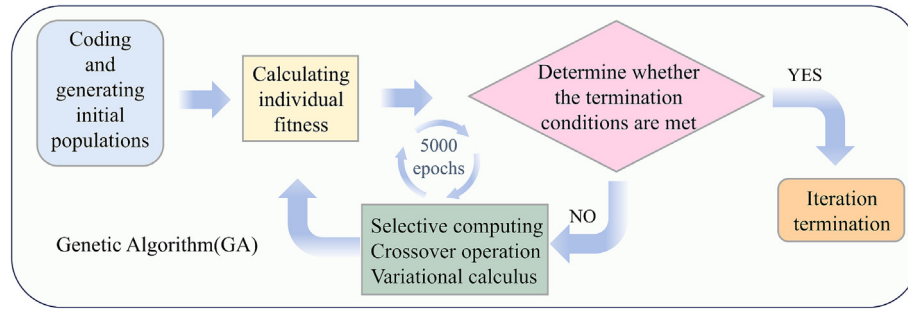


Fig. 2. GA model.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (11)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (12)$$

Where y_i is the actual value, $\hat{y}_i$ is the model estimate, and $\bar{y}$ is the mean of the actual values.

3. Results

3.1. CND analysis under different nitrogen stress conditions

The spectral reflectance characteristics of winter wheat at different growth stages over two years and the corresponding CND distributions are shown in Fig. 4. As shown in Fig. 4a, the spectral curves of winter wheat in the first year exhibited similar overall patterns during the jointing, heading, and flowering stages. However, the reflectance significantly increased during the grain-filling stage, particularly in the visible and near-infrared bands. This increase resulted primarily from leaf senescence, decreased chlorophyll content, and cellular structure degradation. In contrast, compared with the first year, the second year

(Fig. 4b) generally presented higher spectral reflectance levels, with more pronounced differences in the red-edge region (700 to 750 nm), reflecting the impact of varying environmental conditions on plant physiological states across years. Furthermore, the distribution characteristics of CND (Fig. 4c) revealed that the mean CND values and dispersion across the four growth stages differed between the two years. The dispersion was greater during the heading and flowering stages, whereas the CND values were generally lower during the grain-filling stage, indicating that nitrogen accumulation tended to stabilize during this period. The statistical results in Table 3 further corroborate the impact of annual variation on winter wheat: On average, the mean values of chemical indicators during the jointing and heading stages were higher in Y1 (first year) than in Y2 (second year). However, during the flowering and grain filling stages, the mean values in Y2 surpassed those in Y1, precisely aligning with the stage-specific differences in spectral reflectance observed over the two years. Regarding dispersion, the standard deviation for each growth stage in Y1 was greater than that in Y2, matching the higher variability observed for Y1 in the CND scatter plot. In terms of numerical range, the maximum and minimum values of chemical indicators across growth stages in Y1 were generally wider than those in Y2, further illustrating how annual environmental variations regulate the physiological and biochemical status of winter wheat. Overall, winter wheat in both years exhibited pronounced

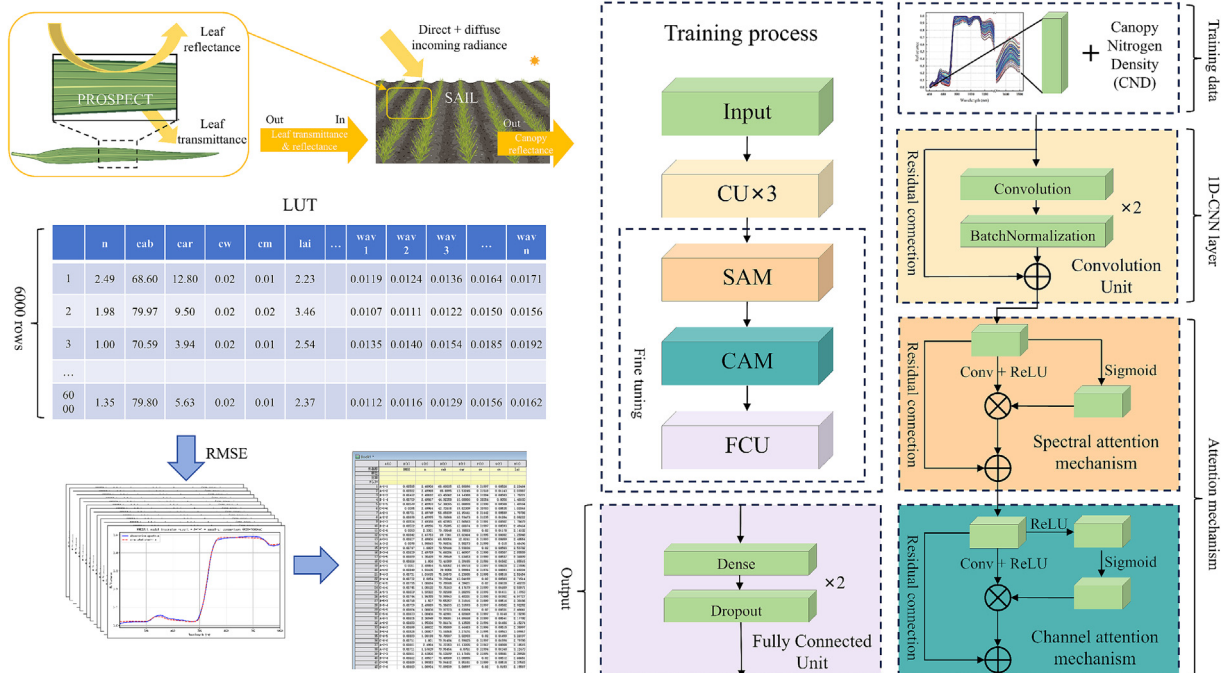


Fig. 3. Framework of the N-PROSAIL radiative transfer model (left) and the SCAR-CNN winter wheat CND estimation model (right).

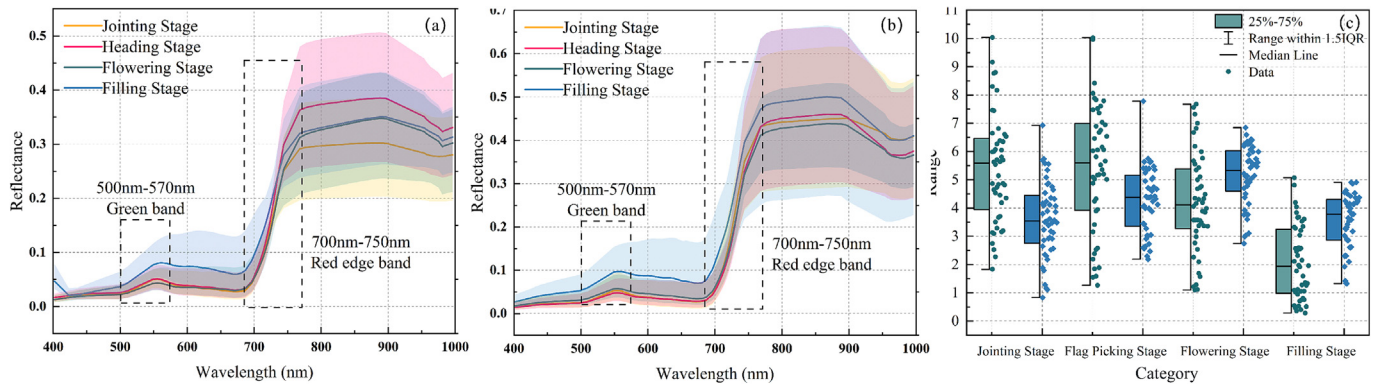


Fig. 4. Spectral and nitrogen density distribution characteristics of winter wheat canopies across different years (a and b show spectral data collected in the first and second years, respectively; c presents a partial box plot and partial scatter plot of nitrogen density for winter wheat across both years; green and blue represent nitrogen density in winter wheat during the first and second years, respectively).

seasonal and interannual variations in spectral and CND indices, providing robust data support for subsequent nitrogen inversion model development.

3.2. Genetic algorithm-based sensitive band selection and validation

To increase the accuracy of the hyperspectral estimation for winter wheat CND, this study employed genetic algorithms (GAs) for sensitive band selection. As an optimization method simulating natural evolutionary processes, genetic algorithms efficiently search for optimal feature combinations within high-dimensional data, making them particularly suitable for eliminating redundant bands and selecting features in hyperspectral datasets. By analyzing the hyperspectral reflectance data of winter wheat canopies, the genetic algorithm screened bands within the 400–1000 nm range, ultimately identifying several sensitive bands located in the visible and near-infrared regions (as shown in Fig. 5a). These bands are closely related to vegetation nitrogen absorption characteristics and canopy structure and are strongly physiologically significant. To validate the reliability of the selected bands, this study further employed the PROSAIL radiative transfer model for forward simulation and inversion verification. The sensitive bands screened by the genetic algorithm were used as inputs to construct a CND estimation model, which was then fitted against the measured data. The results (Fig. 5b) demonstrate that the estimation model based on the GA-selected bands has a high fitting quality, with an R^2 value of 0.98 and an RMSE value of 0.02, indicating excellent agreement between the estimated and measured values. The figure also presents 95% confidence bands and prediction bands, further confirming the model's robust stability and predictive ability.

3.3. CND estimation via RTM, deep learning, and transfer learning

In this study, RTM, deep learning, and transfer learning methods were employed to estimate the CND in winter wheat leaves, and the estimation accuracies of the different methods were compared (Fig. 6). The results from the first year using the benchmark PLSR model for

inversion yielded an R^2 value of 0.77 and an RMSE value of 0.19 (Fig. 6a). The results from the first year using the LUT method for inversion are shown in Fig. 6b, with an R^2 value of 0.45 and an RMSE value of 0.39. Although the model captured some trends, the prediction accuracy was relatively low, indicating poor fit when complex nonlinear relationships were considered. The results from the SCAR model for the first year are shown in Fig. 6c, with a significant improvement in R^2 value to 0.78 and a reduction in the RMSE value to 0.24, demonstrating the advantage of deep learning in capturing complex features. The data in Fig. 6d further illustrate the performance of DTL inversion for the first year, with R^2 value increasing to 0.81 and the RMSE value decreasing to 0.14. This finding demonstrates that transfer learning leverages prior knowledge to further enhance the model's predictive accuracy, highlighting its potential in small-sample learning. The results of the benchmark PLSR model inversion for the second year are shown in Fig. 6e, yielding an R^2 value of 0.49 and an RMSE value of 0.15. The inversion results from the LUT method in the second year are shown in Fig. 6f, with an R^2 value of 0.24 and an RMSE value of 0.23, indicating a decrease in accuracy compared with the lookup table inversion in the first year. The inversion results from the SCAR model in the second year are shown in Fig. 6g, with an R^2 value improving to 0.66 and an RMSE value decreasing to 0.07, demonstrating its ability to handle complex data relationships. The inversion results from the DTL model in the second year are shown in Fig. 6h, with an R^2 value of 0.68 and an RMSE value of 0.07. These results slightly outperformed those of the deep learning inversion, indicating that DTL maintained good accuracy in the second-year application. The results above demonstrate that both the deep learning and DTL methods outperformed lookup table inversion in terms of retrieval accuracy for both years, with the DTL model exhibiting particularly high predictive accuracy in both years. These findings indicate that DTL significantly enhances model predictive capability and stability when diverse data sources are integrated.

3.4. Estimating CND across growth stages and nitrogen gradients

To further validate the applicability of the DTL model across different growth stages and nitrogen levels, this study analyzed CND estimates across growth stages and nitrogen gradients (Fig. 7). Fig. 7(a–h) illustrates the model's predictive performance across different growth stage combinations. The results indicate that when fine-tuned using data from partial growth stages and tested at the remaining stages, the model consistently demonstrated good fitting performance, with R^2 values ranging from 0.43 to 0.89 and RMSE values between 0.12 and 0.25. The prediction accuracy was greater during the heading, flowering, and grain-filling stages than during the jointing stage, indicating that DTL is more sensitive to nitrogen accumulation during periods of relatively stable canopy structure and physiological state.

Table 3
Statistical description of chemical analysis results.

	Jointing Stage		Heading Stage		Flowering Stage		Filling Stage	
	Y1	Y2	Y1	Y2	Y1	Y2	Y1	Y2
Average value	5.41	3.60	5.42	4.27	4.18	5.17	2.06	3.54
Standard deviation	1.92	1.34	2.22	1.16	1.74	1.06	1.31	0.99
Maximum value	10.04	6.93	10.03	7.78	7.68	6.85	5.07	4.91
Minimum value	1.84	0.83	1.27	2.19	1.10	2.75	0.28	1.33

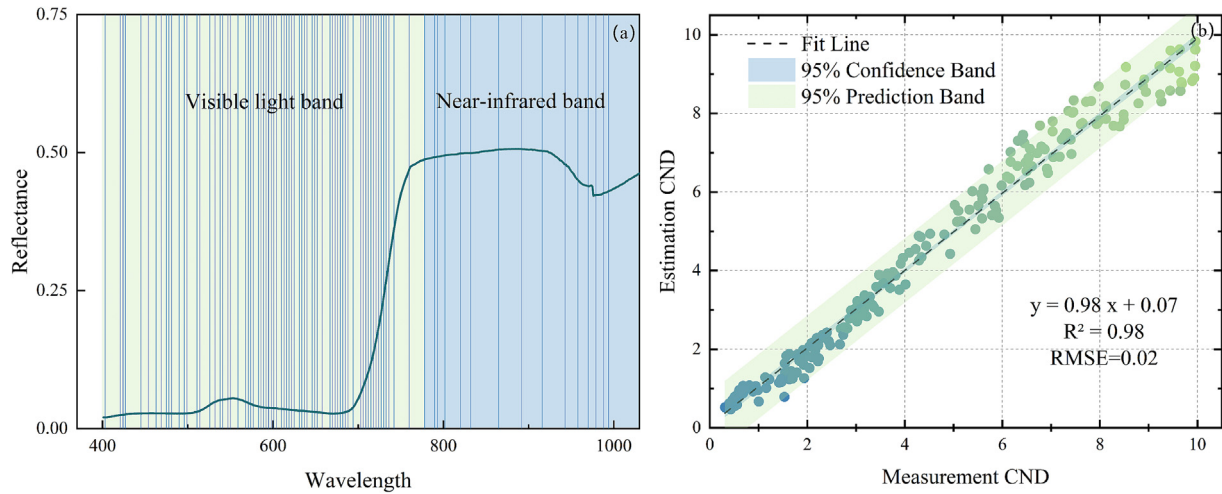


Fig. 5. Distribution of CND-sensitive bands for winter wheat (a) and the N-PROSAIL model validation scatter plot (b).

Additionally, the results from both years demonstrated that models trained across growth stages exhibited strong generalization capabilities, partially overcoming the limitations of single-stage modeling. The model's predictive performance across nitrogen gradients is shown in Fig. 7(i–p). When fine-tuned on three nitrogen gradients and tested on the remaining gradients, the model achieved comparable estimation accuracy, with R^2 values ranging from 0.43 to 0.88 and RMSE values ranging from 0.13 to 0.25. A comparison of the results across nitrogen levels revealed slightly lower estimation accuracy under the low- and high-nitrogen treatments, whereas the predictions remained relatively stable at medium nitrogen levels. These findings indicate that DTL remains robust across a wide range of nitrogen variations but remains sensitive to extreme nitrogen conditions. The DTL model demonstrated strong adaptability and generalization capabilities for estimating CND across growth stages and nitrogen gradients. These results indicate that models trained with integrated information from different growth stages and nitrogen gradients not only increase estimation accuracy but also improve application potential under complex field conditions, providing effective technical support for nitrogen nutrient management in winter wheat.

3.5. Applications to drones and annual transferability

To achieve efficient monitoring of winter wheat nitrogen nutrition status, this study applied a DTL model trained on ground-based hyperspectral data to estimate CND from UAV hyperspectral imagery. In terms of model accuracy, the DTL model effectively captured spatial patterns of nitrogen distribution within the crop canopy. The estimated R^2 values for CND at the jointing, heading, flowering, and grain-filling stages were 0.44, 0.58, 0.56, and 0.68, respectively, with corresponding RMSE values of 0.16, 0.13, 0.22, and 0.15, respectively (see Fig. 8). The model exhibited the highest accuracy during grain filling, indicating the enhanced adaptability of the DTL model during periods of relatively stable leaf morphology and spectral response. The spatial distribution results from the UAV imagery (Fig. 8) further revealed the spatial heterogeneity of the nitrogen distribution in the field. The model estimates clearly distinguished differences under various nitrogen application gradients, showing a gradual increase from low-nitrogen zones to high-nitrogen zones, which is consistent with ground-truth measurements. These findings demonstrate that the DTL model not only achieves high-precision nitrogen estimation at the point scale but also

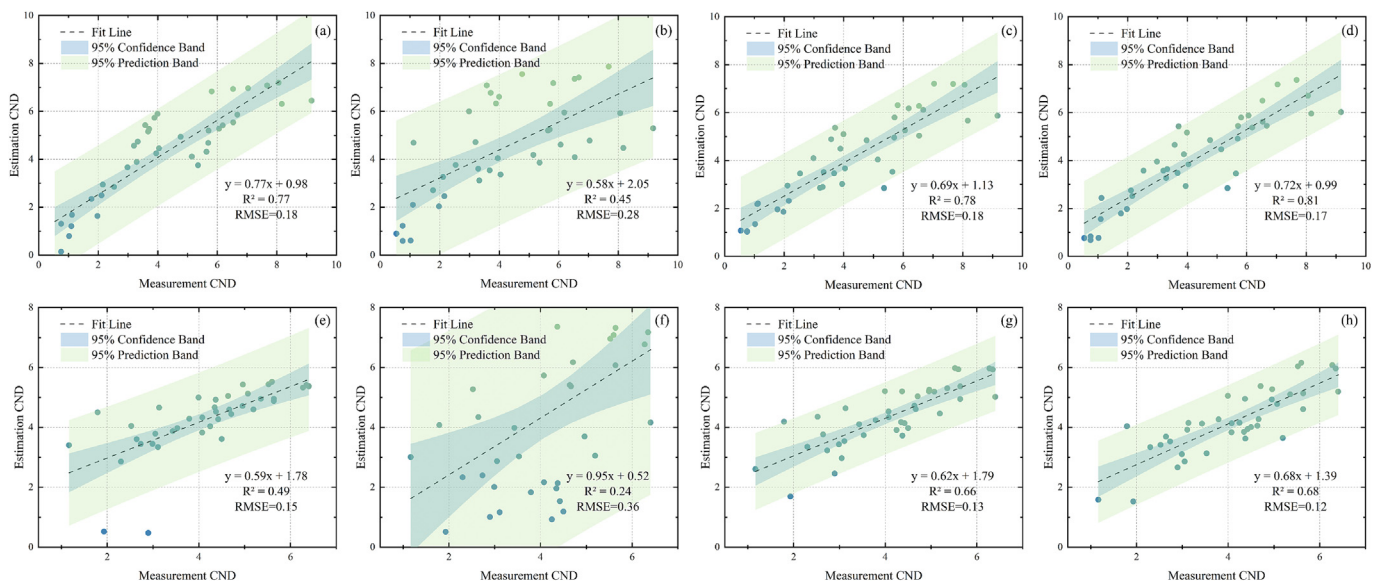


Fig. 6. Model inversion accuracy (Panels a, b, c, and d show the inversion results from the PLSR, RTM, SCAR-CNN, and DTL models for the first year, respectively. The inversion results from the PLSR, RTM, SCAR-CNN, and DTL models for the second year are shown in e, f, g, and h, respectively).

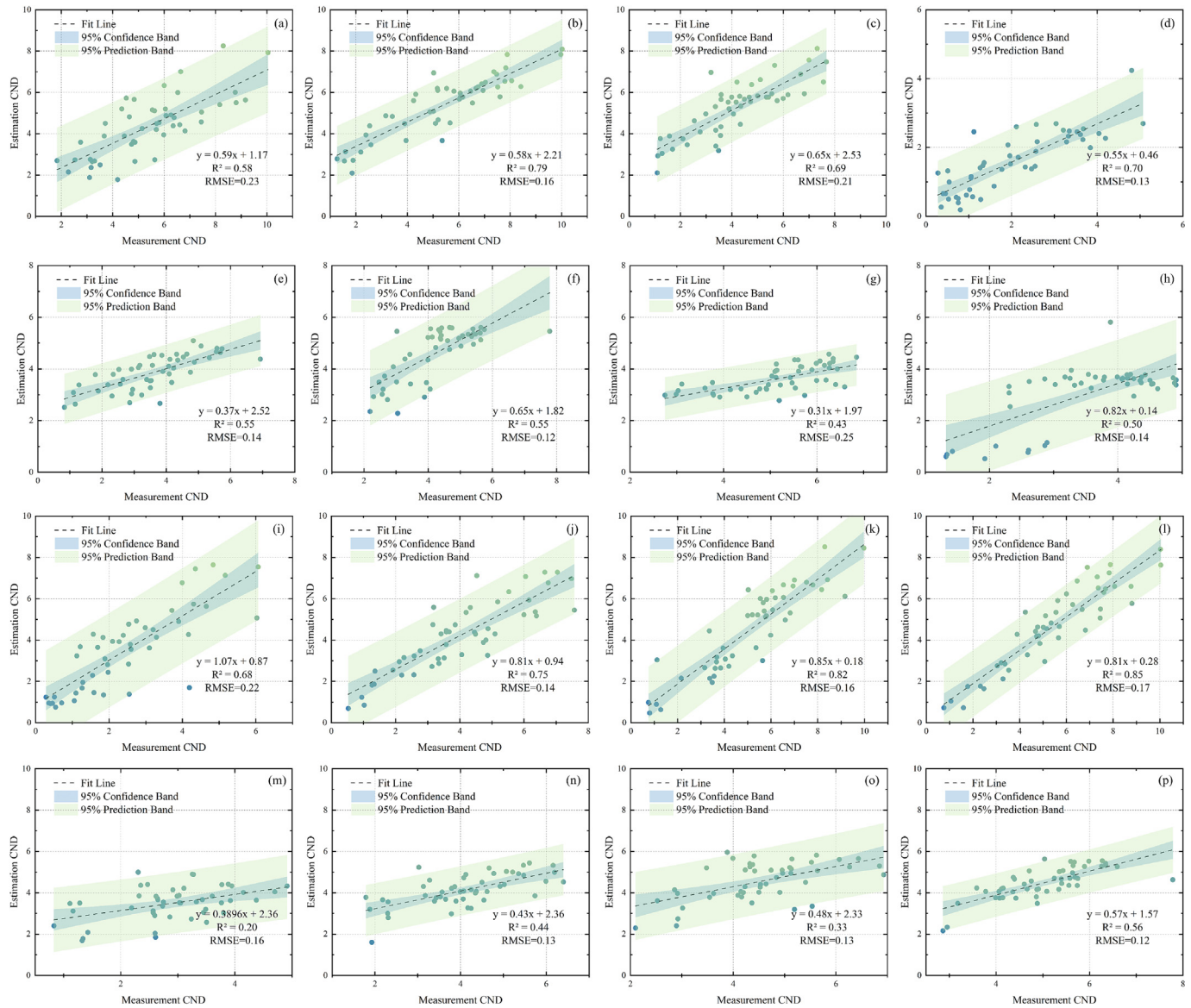


Fig. 7. Inversion accuracy across growth stages and nitrogen gradients (in the figure, a, b, c, and d represent the test set inversion results for the jointing stage, flag-picking stage, flowering stage, and filling stage in the first year, respectively. e, f, g, and h represent the test set inversion results for the jointing stage, flag-picking stage, flowering stage, and filling stage in the second year, respectively. The test set inversion results for nitrogen concentrations of 0, 1, 2, and 3 in the first year are shown in Panels i, j, k, and l, respectively. The test set inversion results for nitrogen concentrations of 0, 1, 2, and 3 in the second year are shown in Panels m, n, o, and p, respectively).

accurately reflects the nitrogen nutrition level of the entire crop population at the regional scale, providing reliable data support for field nitrogen diagnosis and precision management.

To validate the stability and generalization capability of the DTL model in cross-annual applications, this study conducted an annual transferability analysis. The results demonstrate that the model exhibited clear transferability across different years (Fig. 9). When fine-tuned using one year of data and tested on the remaining years, it achieved an R^2 value of 0.50 and an RMSE value of 0.27, demonstrating the robust predictive performance of the model on nonidentically distributed data. The core challenge in cross-annual generalization lies in differences in climatic conditions, management practices, and spectral feature distributions across years. This study confirms that DTL can effectively adapt to these variations, representing a key advantage over traditional methods. This technique significantly mitigates the impact of interannual sample differences on model stability, ultimately achieving usable estimation accuracy in the target year. This provides a reliable

and innovative technical solution for cross-annual monitoring of the nitrogen status of winter wheat.

4. Discussion

4.1. Nitrogen inversion methods

In this study, the performance of LUT, SCAR-CNN, and the DTL model, which combines physical mechanisms with DTL, for estimating the CND of winter wheat was systematically compared. The LUT method, constructed on the basis of the PROSAIL model, has a solid physical foundation but has a low inversion accuracy: $R^2 = 0.45$ and $RMSE = 0.39$ in the first year, and $R^2 = 0.24$ and $RMSE = 0.23$ in the second year. Physical models can reveal in depth the mechanisms of light-vegetation interactions, but their inherent ill-posed inversion problems and sensitivity to input parameter errors and background noise severely limit their practical application in complex agricultural

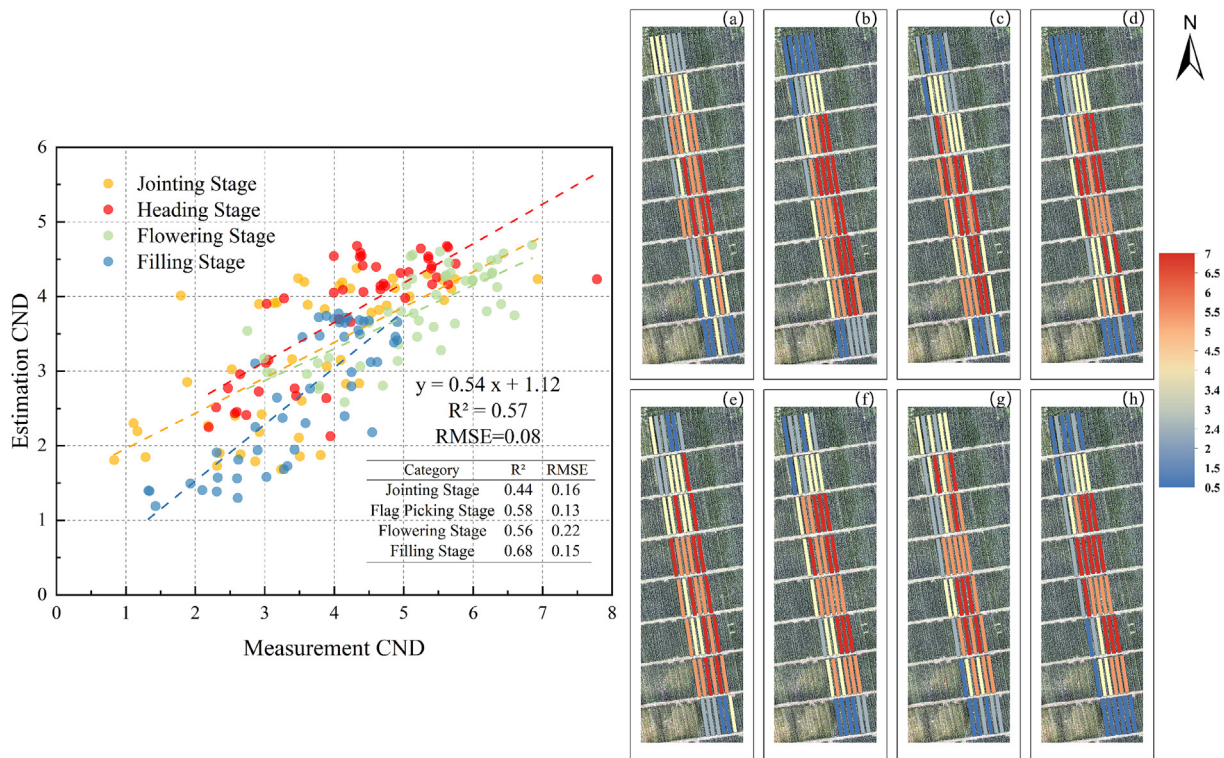


Fig. 8. Scatter plot and image map of drone data inversion (Figures a, c, e, and g represent the measurement CNDs for the Jointing Stage, Heading Stage, Flowering Stage, and Filling Stage in the second year. Figures b, d, f, and h represent the estimation CNDs for the corresponding growth stages).

environments (Li et al., 2018; Sun et al., 2022). In contrast, the SCAR-CNN model demonstrated superior performance in both years: $R^2 = 0.78$ and $RMSE = 0.24$ in the first year and $R^2 = 0.66$ and $RMSE = 0.07$ in the second year. These findings indicate that deep learning has advantages in capturing the complex nonlinear relationships between spectral data and nitrogen density. However, the model's performance remains constrained by the training data volume and distribution, which significantly fluctuates during testing: R^2 values decreased from 0.78 to 0.66. This reflects that feature associations learned by data-driven methods are inherently statistical rather than mechanistic, making them susceptible to environmental variability (Wang et al., 2020). Similarly, vegetation index-based methods often exhibit limited robustness because of their dependence on specific datasets, resulting in

constrained adaptability across years and environmental conditions (Feng et al., 2016; He et al., 2017).

The proposed DTL model demonstrated optimal performance in same-year estimations: the R^2 value was 0.81, with an RMSE value of 0.14 in the first year; and the R^2 value was 0.68, with an RMSE value of 0.07 in the second year. It also demonstrated outstanding robustness and generalization capabilities across growth stages and nitrogen gradients, with R^2 values ranging from 0.43 to 0.89 and 0.43 to 0.88, respectively. By integrating physical mechanisms with DTL, the model significantly outperforms traditional physics-driven and purely data-driven approaches in estimation accuracy, robustness, and generalizability (Chakraborty et al., 2025). Furthermore, this study transferred a DTL model trained on ground data to a drone remote sensing platform, achieving scalability from the point to the field scale. Evaluations on UAV hyperspectral imagery demonstrate progressively improved estimation accuracy across the jointing, heading, flowering, and grain-filling stages, with R^2 values increasing from 0.44 to 0.68. This finding validates the robust scalability of the physical-data dual-driven model (Wang et al., 2023). These results also indicate that when the crop canopy structure stabilizes, the response relationship between the spectral features and nitrogen content becomes more pronounced (Jay et al., 2017; Yao et al., 2017).

4.2. Potential influencing factors

In this study, several potential influencing factors were identified during the construction and validation of the hybrid inversion model. Uncertainties exist in the parameterization process of the radiative transfer model (RTM): if ecological correlations between physiological parameters are ignored during parameter sampling (e.g., unrealistic combinations such as a high leaf area index paired with extremely low chlorophyll content), unreasonable spectral simulation data may be generated, thereby disrupting the training effectiveness of machine learning models (Li et al., 2015). Moreover, relying solely on regional

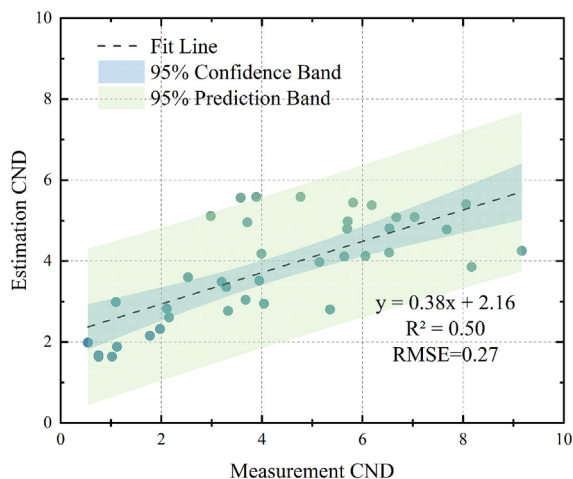


Fig. 9. Interannual inversion accuracy.

or growth stage averages for model parameters while ignoring treatment-specific variations introduces systematic errors (Berger et al., 2018). Therefore, imposing constraints during parameter sampling and optimizing parameter ranges using field observation data are essential. This study also revealed that transfer learning achieved moderate predictive accuracy in the annual transfer scenario ($R^2 = 0.50$; RMSE = 0.27). This phenomenon stems from spectral morphological differences caused by interannual climatic conditions and field management practices. When transferable models are constructed, fully accounting for the effects of environmental variability and domain shift is essential (Wan et al., 2020). Additionally, remote sensing data sources and spatial scales warrant attention: Although hyperspectral data offer significant advantages in feature recognition, their high acquisition costs and processing complexity limit practical applications. Current satellite-borne hyperspectral sensors (e.g., EnMAP and PRISMA) still exhibit coarse spatial resolution (Berger et al., 2020a; Guanter et al., 2025; Loizzo et al., 2019), limiting their ability to characterize ground features. Data usability must be enhanced through multisource data fusion or downscaling methods.

4.3. Prospects and outlook

Future research may further explore this topic in the following areas. By enhancing the ability of the RTM to characterize three-dimensional canopy structure, soil background, and atmospheric influences (Vicent et al., 2020), the reliability of forward simulations can be improved, advancing the coupling of RTMs with crop growth models (CGMs) to jointly simulate nitrogen accumulation, distribution, and turnover processes (Thorp et al., 2012). This will enable more accurate inversion of crop biochemical parameter dynamics. Such integrated models play a critical role in precision nitrogen fertilizer management under changing environmental conditions. With the development of next-generation hyperspectral satellite constellations, conditions are now ripe for establishing an integrated “ground-UAV-satellite” multiscale validation and inversion system. Key efforts should focus on exploring data fusion mechanisms between hyperspectral and multispectral data (e.g., Sentinel-2 and Landsat-8), developing hybrid inversion workflows adaptable to varying resolution requirements, and achieving a balance between cost and accuracy to advance operational applications of large-scale agricultural remote sensing monitoring (Verrelst et al., 2021).

5. Conclusions

In this study, a hybrid framework integrating radiative transfer modeling with DTL for remote sensing estimation of winter wheat CND was successfully developed and validated. The proposed physics-driven and data-driven deep fusion framework fully leverages the generalization ability of the N-PROSAIL model in generating physically constrained data and the nonlinear feature extraction capability of deep learning. Through transfer learning strategies, this framework achieves high-precision CND estimation with minimal field samples for fine-tuning, significantly outperforming traditional LUT inversion methods and purely data-driven deep learning models. This resolves the core challenge of model training and generalization under limited sample conditions. The model maintains high estimation accuracy (R^2 values range from 0.43 to 0.89) and robustness across different growth stages and nitrogen fertilizer treatment gradients, demonstrating excellent adaptability to complex field conditions. Further application to drone hyperspectral imagery successfully achieved spatial scaling from the point to the field scale, with R^2 values reaching 0.68 during the grain-filling stage, demonstrating the practical potential of the method for precision agriculture variable-rate fertilization. When cross-year transferability was evaluated, the model was fine-tuned using one year of data and tested on the remaining years ($R^2 = 0.50$). This revealed challenges to the model's generalization ability stemming from shifts in data distribution caused by environmental variability and

differences in management practices across years. This study not only provides a novel and reliable technical solution for the nondestructive monitoring of nitrogen nutrition in winter wheat but also offers an innovative research paradigm that integrates physical mechanisms with data-driven approaches to address model transferability and small-sample learning challenges in agricultural remote sensing. It holds significant theoretical value and promising application prospects.

CRedit authorship contribution statement

Bo Yang: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation. **Longfei Zhou:** Writing – review & editing, Supervision, Project administration, Data curation. **Guijun Yang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Guangsheng Zhang:** Writing – review & editing, Supervision, Resources, Investigation. **Jingwei Qi:** Writing – review & editing, Supervision, Resources, Project administration. **Changchun Li:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

Guijun Yang is an editor for Artificial Intelligence in Agriculture and was not involved in the editorial review or the decision to publish this article. All authors declare that there are no competing interests.

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